

Multi-Battery Swapping Station Modelling for EVs With Strategic Discounting and Solar Energy

Manish Sharma and Deep Kiran

Department of Electrical Engineering, Indian Institute of Technology Roorkee, Roorkee, Uttarakhand, India

Email: conveytomanish@gmail.com; deepkiran@ee.iitr.ac.in

Abstract—The successful roll-out of electric vehicles (EVs) in the market requires adequate charging infrastructure to mitigate concerns regarding their range and long charging delays. With this motivation, a business model is proposed in this paper for the modelling of multi-battery swapping stations for EVs with some strategic discounts. The proposed model generates profits for their owners while providing an efficient alternative to charging EV batteries. This model also utilizes solar energy for charging these batteries, with the provision to supply excess power back to the grid. The proposed model considers two different types of batteries along with discounts during solar peak hours. The results show its feasibility and effectiveness in providing the required services.

Index Terms—Battery swapping stations, electric vehicles, solar energy, and strategic discounts.

NOMENCLATURE

Binary variables:

$a_{i,g,t}$	Auxiliary variable for battery i of group g at time t .
c_t	Auxiliary variable at time t .
u_t	Binary variable for PV peak hours time (1 during PV peak hours and 0 otherwise).
$x_{i,g,t}$	Binary variable for swapping status of battery i of group g at time t .

Continuous variables:

$\beta_{i,g,t}$	Discount percentage offered to the customer exchanging the battery i of group g at time t .
$batt_{i,g,t}^{chr}$	Charging power of battery i of group g during time t (in kW).
$batt_{i,g,t}^{dis}$	Discharging power of battery i of group g during time t (in kW).
el_t^{buy}	Electricity imported from the wholesale market at time t .
el_t^{sell}	Electricity exported to the wholesale market at time t .
hh_t	PV peak hours discount offered at a particular time t .
hhc_t	Cost incurred by giving PV peak hours discounts.
$J_{i,g,t,d}$	Energy storage of battery i of group g at time t in the discount segment d on the discount curve (in kWh).

pv_t^{in}

Input solar energy at time t .

$SC_{i,g,t}$

State of charge of battery i of group g in time t (in kWh).

$SC_{i,g,t}^{shrt}$

Shortage of energy of battery i of group g at time t when it was being swapped.

Integer variables:

$batt_{g,t}^{shrt}$

Batteries shortage of group g at time t .

Parameters:

η_{char}

Efficiency of battery charging.

η_{dis}

Efficiency of battery discharging.

λ_{DA}^t

Predicted DA market price at time t .

$BCap_g$

Theoretical battery capacity of group g .

$BCap_g^{max}$

Maximum battery capacity of group g (in kWh).

$BCap_g^{min}$

Minimum battery capacity of group g (in kWh).

BSR_g

Earnings received from the customers for a battery exchange for group g .

CD

Customer dissatisfaction quotient.

J_d^{max}

Maximum limit of each state of charge shortage segment d .

M

Large integer value.

m_d

Discount curve slope in segment d (%/kWh).

$Nbat_{g,t}$

Predicted demand of battery of group g at time t .

$P_{i,g}^{max}$

Maximum power of battery i of group g .

pv_{max}^{in}

Maximum energy generated by renewable energy source.

$SC_{i,g,t}^{init}$

Initial state of charge of battery i of group g at time t .

I. INTRODUCTION

Electric vehicles (EVs) are fully powered on electric power, having fewer running costs due to lesser moving parts for maintenance. Moreover, they are eco-friendly due to their less reliance on fossil fuels. As a result, they are expected to be the future alternative to conventional fossil fuel based-vehicles. EV adoption has spread worldwide in many categories of vehicles, such as trucks, buses, cars, and two-wheelers [1]. Also, technological advancements have supported their transformation over the past few decades [2], [3]. However, their successful roll-out into the market is still held back due to limited charging points, short driving range, and long recharging times [4], [5].

A battery swapping station (BSS) can address these limitations to a large extent. Tesla Motors introduced battery swapping technology for their EVs [6]. Additionally, many

Chinese utilities deployed BSSs for their EV owners in 2013 [7]. BSS serves the same purpose as a gasoline station for traditional internal combustion engine vehicles. Instead of charging the EVs, BSS facilitates replacing its discharged battery with a charged one at a fixed rate, known as the battery swapping rate (BSR). Additionally, BSS can participate in electricity markets to optimize its power requirements and provide services such as demand response, energy storage and reserves. These abilities of the BSS vary depending on the time-varying electricity prices, i.e., the day-ahead (DA) pricing. Fig. 1 shows the interaction of BSS with EVs, rooftop solar

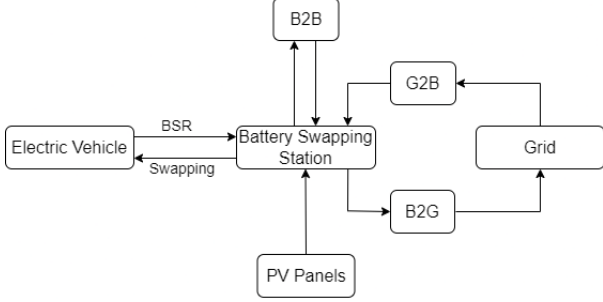


Fig. 1. Interaction between BSS with EVs, PV panels and grid.

(PV) panels and the grid. It can operate in the following three modes. 1) Grid-to-Battery (G2B) mode: imports the energy to charge the batteries during low DA prices. 2) Battery-to-Grid (B2G) mode: exports power back to the grid to ensure profits during high DA prices. 3) Battery-to-Battery (B2B) mode: transfers the power from one or many batteries to charge others. The rooftop PV panels support reducing its dependence on the grid and may make it self-sustainable. The surplus solar energy can also be exported back to the grid, ensuring higher profits. It may additionally offer incentives such as solar peak charging and other strategic discounts, which can enable the commercial viability of its operation.

In the areas of its development and operation, certain pioneering efforts have already been done. In [8], a DA scheduling model of a BSS is presented with one category of EV battery which sets the foundation for this paper. However, it is limited to the BSS owner's viewpoint while neglecting the purview of the customer, such as offering some strategic discounts to retain the customers. In [9], [10], a dynamic pricing strategy is described to increase the revenue of an EV charging infrastructure without considering the customers' experiences, i.e., discounting features. The best places to install and operate BSS in distribution networks are identified in [11]. This model specifically takes into account the type of load, the distribution system's need for reinforcements, and the system's dependability. The EV model, however, employs a heuristic method to choose charging and discharging schedules.

To bridge these gaps, the proposed DA model utilizes the B2G and G2B services for two categories of batteries

for scheduling BSS. This model offers customers two types of strategic discounts for their convenience, profitability and retention. 1) A discount is provided to compensate for the unavailability of fully charged batteries. 2) A discount is provided for swapping during solar peak hours. These features ensure a win-win situation for both BSS and EV owners. With this, the main contributions of this paper are:

- DA scheduling model of a BSS is proposed to take advantage of B2G and G2B services for two categories of batteries. Moreover, it incorporates the effect of varying DA prices from the electricity markets. Additionally, it includes rooftop PV panels to reduce the cost associated with the imported power from the grid.
- This model incorporates strategic discounts for the customers to retain them by discounting for partially charged batteries and swapping during solar peak hours.

This paper is divided into four sections. Section I presents its motivation and a few works of literature in this area. Section II explains the proposed model. A case study with the results and discussion is given in section III. Finally, section IV concludes the study.

II. PROPOSED BUSINESS MODEL FOR BSS

The proposed model facilitates the swapping of two types of batteries at BSS having different characteristics, such as charging/ discharging rates and maximum power capacities. It offers discounts to customers during a swapping of partially charged batteries. It is assumed that the BSS forecasts the DA prices (λ_{DA}^t) and battery swapping demand ($Nbat_{g,t}$) to schedule its operations. The implementation of all these features is explained next.

A. DA model of a BSS

The proposed model maximizes the profits of BSS by optimal scheduling the charging/ discharging of batteries and their power import/ export from the grid to utilize solar energy efficiently. The formulation of this model is shown below:

$$\begin{aligned} \max \quad & \sum_{i \in I} \sum_{t \in T} \sum_{g \in G} x_{i,g,t} BSR_g - \sum_{t \in T} \lambda_{DA}^t (el_t^{buy} - el_t^{sell}) \\ & - CD \sum_{t \in T} \sum_{g \in G} batt_{g,t}^{shrt} - \sum_{i \in I} \sum_{t \in T} \sum_{g \in G} \beta_{i,g,t} BSR_g \end{aligned} \quad (1)$$

subject to:

$$\begin{aligned} SC_{i,g,t} = & (SC_{i,g,t-1} + batt_{i,g,t}^{chr} - \frac{batt_{i,g,t}^{dis}}{\eta_{dis}})(1 - x_{i,g,t}) \\ & + SC_{i,g,t}^{init} x_{i,g,t} \quad \forall t \in T, i \in I, g \in G, \end{aligned} \quad (2)$$

$$SC_{i,g,t-1} + SC_{i,g,t}^{shrt} \geq BCap_g^{max} x_{i,g,t} \quad \forall t \in T, i \in I, g \in G \quad (3)$$

$$\sum_{i \in I} \sum_{g \in G} x_{i,g,t} + batt_{g,t}^{shrt} = Nbat_{g,t} \quad \forall t \in T \quad (4)$$

$$el_t^{buy} - el_t^{sell} = \sum_{i \in I} \sum_{g \in G} (batt_{i,g,t}^{chr} - batt_{i,g,t}^{dis}) \quad \forall t \in T \quad (5)$$

$$0 \leq batt_{i,g,t}^{chr} \leq x_{i,g,t} P_{i,g}^{max} \quad \forall t \in T, i \in I, g \in G \quad (6)$$

$$0 \leq batt_{i,g,t}^{dis} \leq (1 - x_{i,g,t}) P_{i,g}^{max} \quad \forall t \in T, i \in I, g \in G \quad (7)$$

$$BCap_g^{min} \leq SC_{i,g,t} \leq BCap_g^{max} \quad \forall t \in T, i \in I, g \in G \quad (8)$$

$$BCap_g^{min} \leq SC_{i,g,t}^{shrt} \leq BCap_g^{max} \quad \forall t \in T, i \in I, g \in G \quad (9)$$

$$batt_{i,g,t}^{dis} \leq P^{max} a_{i,g,t} \quad \forall t \in T, i \in I, g \in G \quad (10)$$

$$batt_{i,g,t}^{chr} \leq P^{max} (1 - a_{i,g,t}) \quad \forall t \in T, i \in I, g \in G \quad (11)$$

$$el_t^{buy} \leq M c_t \quad \forall t \in T \quad (12)$$

$$el_t^{sell} \leq M (1 - c_t) \quad \forall t \in T \quad (13)$$

$$SC_{i,g,t=|T|} = SC_{i,g,t=0}^{init} \quad \forall i \in I, g \in G \quad (14)$$

$$\frac{SC_{i,g,t}^{shrt}}{BCap_g} = \sum_{d \in D} J_{i,g,t,d} \quad \forall i \in I, t \in T, g \in G \quad (15)$$

$$\beta_{i,g,t} = \sum_{d \in D} m_d J_{i,g,t,d} \quad \forall i \in I, t \in T, g \in G \quad (16)$$

$$0 \leq J_{i,g,t,d} \leq J_d^{max} \quad \forall t \in T, i \in I, g \in G, d \in D \quad (17)$$

The above objective function (1) comprises four terms that collectively determine the BSS's profit. The first term indicates the profit earned from swapping at BSR_g , where $x_{i,g,t}$ is equal to one if swapping takes place and zero otherwise. The second term determines the profit earned from the sale and purchase of electricity from the market at DA prices (λ_t^{DA}). The third term indicates the penalty incurred by the BSS in case of battery shortage using a customer dissatisfaction factor (CD). It represents the loss suffered by BSS due to its inability to provide a battery to the customers. The BSS may calculate CD based on two factors: revenue loss if swapping may have been executed and financial loss from failing to attract new customers. The fourth term shows the discount on battery ($\beta_{i,g,t}$) offered to a customer if BSS cannot offer a 100% charged battery. In (2), the state of charge (SOC) ($SC_{i,g,t}$) of each battery i is calculated considering its previous state ($SC_{i,g,t-1}$), charging ($batt_{i,g,t}^{chr}$) or discharging power ($batt_{i,g,t}^{dis}$), and efficiency of the battery. In order to linearize (2), a Big-M method or method of penalties is used by considering a significantly large value (M) [12]. In order to obtain a basic feasible solution, (2) is linearized as given in (18)-(21). It can be observed that when $x_{i,g,t} = 0$, then the SOC of that particular battery is the same as the previous time slot, and when $x_{i,g,t} = 1$, then the SOC of that specific battery is updated as per new arrived battery.

$$SC_{i,g,t} - (SC_{i,g,t-1} + batt_{i,g,t}^{chr} - \frac{batt_{i,g,t}^{dis}}{\eta_{dis}}) \leq M x_{i,g,t} \quad (18)$$

$$SC_{i,g,t} - (SC_{i,g,t-1} + batt_{i,g,t}^{chr} - \frac{batt_{i,g,t}^{dis}}{\eta_{dis}}) \geq -M x_{i,g,t} \quad (19)$$

$$SC_{i,g,t} - SC_{i,g,t}^{init} \leq M (1 - x_{i,g,t}) \quad (20)$$

$$SC_{i,g,t} - SC_{i,g,t}^{init} \geq -M (1 - x_{i,g,t}) \quad (21)$$

Eq. (3) deals with the mechanics of battery swapping. If battery swapping takes place, then the SOC should be at the maximum battery capacity ($BCap_g^{max}$). If BSS cannot meet the maximum battery capacity, then the energy shortage ($SC_{i,g,t}^{shrt}$) takes account of the remainder. This variable capturing the effect of battery shortage is penalized in (1). The energy shortage requirements can be purchased from the market to fulfil demand, and any surplus can be sold back for additional profits for the BSS, as shown in (5). In (4), the BSS seeks to meet the battery demand ($Nbat_{g,t}$). Eqs. (6) and (7) enforce the constraint for maximum charging/ discharging power limit ($P_{i,g}^{max}$). The SOC and its shortages of batteries are constrained by minimum and maximum capacities in (8) and (9). The charging and discharging of batteries at any given time t are ensured in (10) and (11). The buying and selling of electricity are satisfied in (12) and (13). Eq. (14) denotes the SOC of battery i of group g in $t = T^{th}$ time slot is equal to the initial SOC of the same battery during time slot $t = 0$. If the BSS cannot provide a fully charged battery for swapping, it can incur a CD penalty. The discounts are modelled in the piece-wise linear form given in Fig. 2 in which m_d ($m_1, m_2, m_3, m_4, \dots$) is the slope of each discount segment, and here the offering increases with the increase in battery shortage. It is a penalty for the inconvenience to the customer for a battery swapping having a lower SOC. They are modelled in (15), (16) and (17). Eq. (15) calculates the normalized shortage in the SOC and distributes it in discrete segments ($J_{i,g,t,d}$). In (16), the discount ($\beta_{i,g,t}$) is calculated as the product of the slope of each discount segment m_d and the respective shortage segment J_d . Eq. (17) limits the SOC shortage segments to their preset limits J_d^{max} .

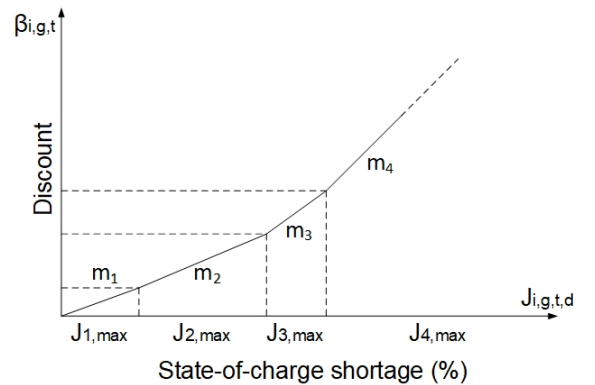


Fig. 2. Piece-wise linear discount function.

B. Inclusion of solar energy

Installing solar panels at the BSS can supplement its energy demand, reducing its dependency on buying power from the

electricity market or grid. Also, the BSS can sell back the surplus power generated from these panels to the electricity markets. This model assumes that solar energy is freely available, ignoring its installation and operational costs distributed over its lifespan. This is included in the model by replacing (5) as:

$$el_t^{buy} - el_t^{sell} + pv_t^{in} = \sum_{i \in I} \sum_{g \in G} (batt_{i,g,t}^{chr} - batt_{i,g,t}^{dis}) \quad \forall t \in T \quad (22)$$

In (22), the difference between solar energy (pv_t^{in}) plus imported energy (el_t^{buy}) and exported energy (el_t^{sell}) signify the net energy delivered by BSS to all the batteries during the entire time interval.

C. Incorporation of strategic discounts during solar peak hours

The BSS can offer an additional discount to customers during solar peak time intervals for better utilization. This is expressed as follows:

$$hh_t = 0.2BSR_g u_t \quad \forall t \in T \quad (23)$$

where,

$$u_t = \begin{cases} 1, & \text{If } pv_t^{in} \geq 0.7pv_{max}^{in} \\ 0, & \text{Otherwise} \end{cases} \quad (24)$$

$$hhc_t = x_{i,g,t} hh_t \quad \forall i \in I, t \in T, g \in G \quad (25)$$

Here, it is assumed that the BSS provides a 20% discount on the BSR_g when solar power (pv_t^{in}) generation is above 70% of its maximum capacity (pv_{max}^{in}) which are modelled using (23) and (24). These values are taken for illustrative purposes and may be adjusted based on the choice of BSS. In (24), solar peak hours discount (hh_t) is offered when the condition of PV peak hours is met for a particular time t . In (25), hhc_t signifies the cost incurred to the BSS in case the given battery is swapped during the PV peak hours. Summation of hhc_t over time gives customers a discount during solar peak hours. This effect is included in the objective function as follows:

$$\begin{aligned} & \sum_{i \in I} \sum_{t \in T} \sum_{g \in G} x_{i,g,t} BSR_g - \sum_{t \in T} \lambda_{DA}^t (el_t^{buy} - el_t^{sell}) \\ & - CD \sum_{t \in T} \sum_{g \in G} batt_{g,t}^{shrt} - \sum_{i \in I} \sum_{t \in T} \sum_{g \in G} \beta_{i,g,t} BSR_g - \sum_{t \in T} hhc_t \end{aligned} \quad (26)$$

The objective function in (1) is replaced by (26) to incorporate discounts for customers swapping during solar peak hours.

III. CASE STUDY AND DISCUSSIONS

In this section, the proposed model is validated for the optimal operation of a multi-battery BSS. The model is implemented in Matlab using Gurobi optimization solver [13] and YALMIP optimizer [14]. This simulation is carried out for a day (24 hours), having 96 time slots of 15 minutes each. Two types of batteries (2-wheelers (2-W) with 12kWh

capacity and 4-wheelers (4-W) with 24kWh capacity), with ten each available for swapping, are used for the simulation. The batteries' charging and discharging efficiency are assumed to be 90%. Here, a 7 kW rooftop solar plant is also used as a source of renewable energy. The profit earned through trade with the grid is calculated using the DA prices in the electricity market, as shown in Fig. 3 [15]. BSR for 2-W and 4-W is taken to be ₹ 70 and ₹ 100, respectively, as taken in [6]. The calculation of CD is beyond the scope of this study. Therefore, for simplicity, it is assumed to be 20 and 30 for 2-W and 4-W batteries, respectively. Although, a BSS can make an informed decision on the choice of CD based on their unmet demand. In this study, the value of m_d is presumed to be 0.8. The power received from the solar panels at the BSS over the entire day is shown in Fig. 4. It can be seen that a maximum solar power output (pv_{in}^{max}) is observed at around 48th slot (about noon). The EVs arriving at the BSS for battery swapping and

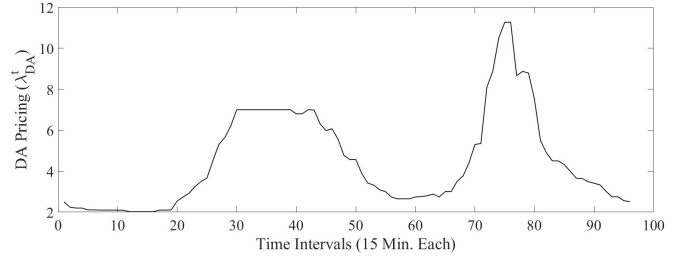


Fig. 3. DA prices (in ₹/kWh) gathered from Indian Energy Exchange [15].

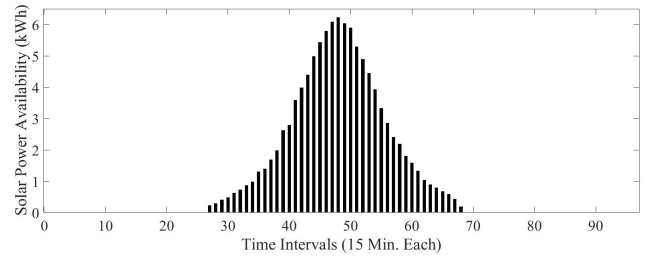


Fig. 4. Solar power available during the day.

their SOC are unknown; therefore, some random values are assumed for illustration purposes. This assumption is justifiable since no historical data is available for any current BSS [8]. It is presumed that all incoming batteries have a charge of 50% or less. The initial SOC of all 2-W and 4-W batteries arriving throughout the time intervals is shown in Figs. 5 and 6, respectively. It can be observed that, in each time interval, 10 EVs of each type are arriving for battery swapping. Thus, a total of 1920 battery (960 for each 2-W and 4-W) swapping requests are received for a day by this BSS. Three cases are simulated based on the above information to demonstrate the proposed model in section II.

1) *DA case (C1)*: The optimization model for a DA case (C1) is given as follows. Max

$$\sum_{i \in I} \sum_{t \in T} \sum_{g \in G} x_{i,g,t} BSR_g - \sum_{t \in T} \lambda_{DA}^t (el_t^{buy} - el_t^{sell}) - CD \sum_{t \in T} \sum_{g \in G} batt_{g,t}^{shrt} - \sum_{i \in I} \sum_{t \in T} \sum_{g \in G} \beta_{i,g,t} BSR_g \quad (27)$$

subject to, Constraints (2) - (21).

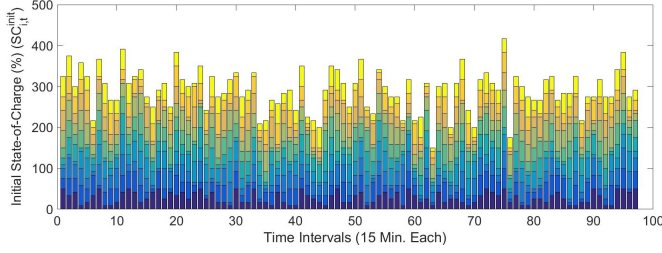


Fig. 5. Initial state of charge of 2-W batteries.

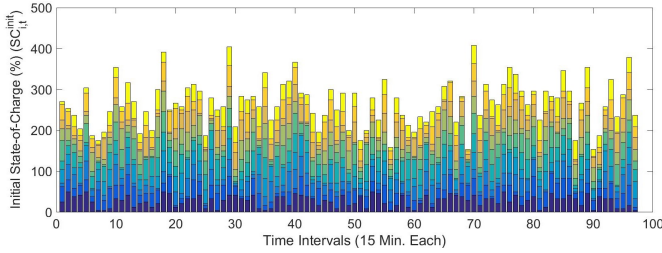


Fig. 6. Initial state of charge of 4-W batteries.

2) *C1 with solar energy (C2)*: In addition to C1, this case includes the effect of solar energy. The optimization model for this case is given as follows. Max

$$\sum_{i \in I} \sum_{t \in T} \sum_{g \in G} x_{i,g,t} BSR_g - \sum_{t \in T} \lambda_{DA}^t (el_t^{buy} - el_t^{sell}) - CD \sum_{t \in T} \sum_{g \in G} batt_{g,t}^{shrt} - \sum_{i \in I} \sum_{t \in T} \sum_{g \in G} \beta_{i,g,t} BSR_g \quad (28)$$

subject to, Constraints (2) - (22).

3) *C2 with solar peak hours discounts (C3)*: In addition to C2, this case includes the effect of solar peak hour discounting for EVs. The optimization model for this case is given as follows. Max

$$\sum_{i \in I} \sum_{t \in T} \sum_{g \in G} x_{i,g,t} BSR_g - \sum_{t \in T} \lambda_{DA}^t (el_t^{buy} - el_t^{sell}) - CD \sum_{t \in T} \sum_{g \in G} batt_{g,t}^{shrt} - \sum_{i \in I} \sum_{t \in T} \sum_{g \in G} \beta_{i,g,t} BSR_g - \sum_{t \in T} hh c_t \quad (29)$$

subject to, Constraints (2) - (25). Based on the above information, the total number of swapping done by this BSS

is 1355, of which 683 and 672 batteries are 2-W and 4-W, respectively. The swapping taking place are based on the charged batteries available in the BSS at that particular time period. The SOC of the 2-W and 4-W batteries available to swap throughout the time period are given in the Fig. 7 and Fig. 8 respectively with the number of batteries swapped in all three cases. From these figures it can be observed that when the quantity of charged batteries is high, consequently the rate of battery swapping is also high. In the above three models,

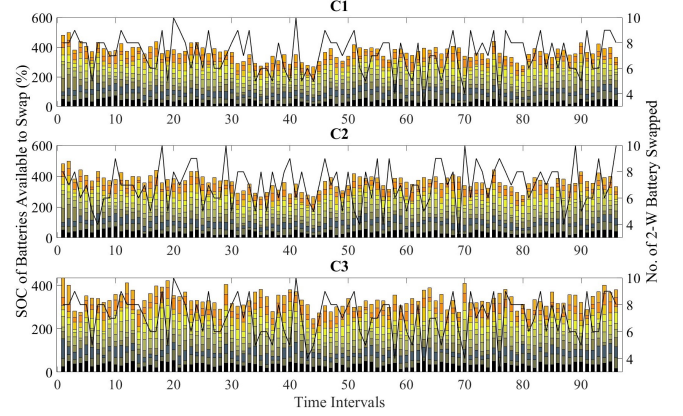


Fig. 7. SOC of the 2-W batteries available to swap throughout the study period.

the profits obtained are ₹ 46,390.8473, ₹ 46,920.3839 and ₹ 44,408.9518 per day in C1, C2 and C3, respectively. It can be observed that the consideration of solar energy as a free energy source has boosted the profits in C2 compared to C1. Due to the implementation of strategic solar peak hour discounting, a portion of the BSS's profit is shared among EVs as an incentive for swapping during these times. Such modelling is expected

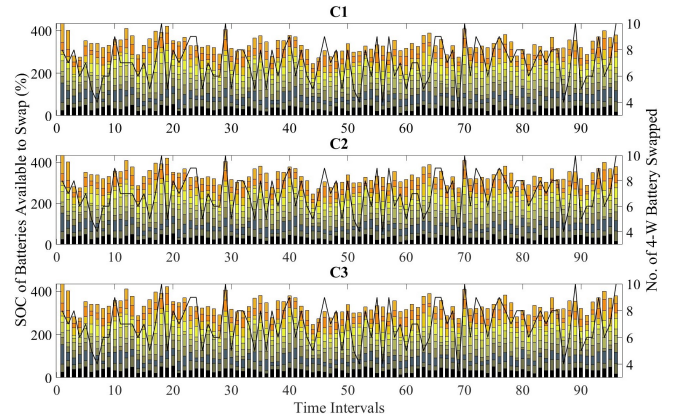


Fig. 8. SOC of the 4-W batteries available to swap throughout the study period.

to assist the BSS in retaining customers by sharing a portion of its profits. Therefore, the profit in C3 is lesser than that of C2. The discount offered to the customers during peak hours is

given in Fig. 9. The energy imported from (+ve) and exported (-ve) to the electricity market by the BSS is given in Table I. In C2, the imported energy has reduced compared to C1 due to the inclusion of the free solar energy source. In C3, the energy import remains almost the same as it only rearranges EV charging during solar peak hours. From Fig. 10, it can be

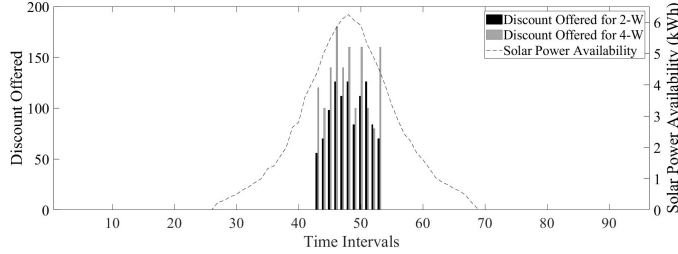


Fig. 9. Discount offered (in ₹) in solar peak hours throughout the time intervals.

TABLE I

ENERGY IMPORTED (+VE) AND EXPORTED (-VE) BY THE BSS IN EACH CASE (IN KWH).

	C1	C2	C3
Imported	411.3558	375.9999	374.3399
Exported	-215.7609	-291.709	-288.93

observed that the stored energy is sold back to the grid during high-price time intervals which were stored during low-price time intervals. This supports the generation of profits for BSS. Here, the BSS prefers exporting energy to the grid instead of charging of batteries and offering them at a discount. In C3, the

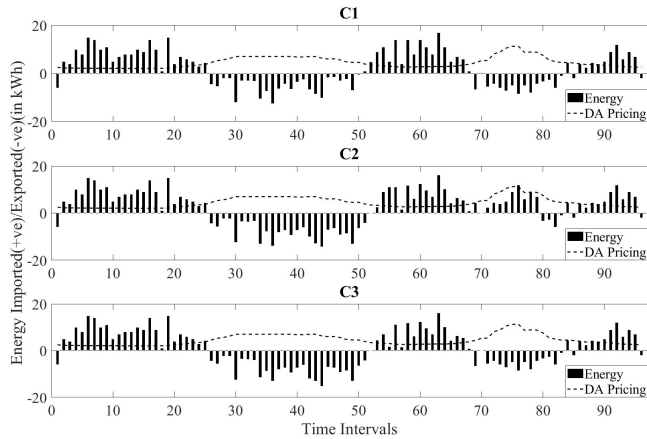


Fig. 10. Variation of energy imported/exported to electricity market over time horizons for each case.

cost corresponding to the CD terms in (29) is ₹ 14,180. Due to the partially charged battery swaps, the discount offered to customers is ₹ 54,865. The strategic discount offered for charging during solar peak hours is ₹ 2511.4321. Based on the above costs, it can be observed that a BSS has opportunities to

offer various economic flexibility for the customers. Moreover, it can ensure its profits by continuously monitoring these costs and taking appropriate actions to improve them.

IV. CONCLUSION

This paper proposes a business model for the optimal operational scheduling of a BSS with multiple battery types. The proposed model is simulated with DA prices from electricity markets and incorporating rooftop solar panels at the BSS, ensuring more profits. Two types of discounts are offered to the customers for swapping their EV batteries. One is for swapping of partially charged batteries, and another for swapping at solar peak hours. These discounts reduce its profits and support in shifting the peak demand to off-peak periods. Additionally, it can offer ancillary support services to the grid. The proposed model can help stakeholders to operate and design the BSS. It will also allow more efficient market decisions that will help utilize the BSS capabilities and enhance environmental sustainability by introducing solar energy.

REFERENCES

- [1] J. Cobb, "Global plug-in car sales cruise past 1.5 million," HybridCars.com, Toronto, ON, Canada, Jun. 22, 2016. [Online]. Available: <http://www.hybridcars.com/global-plug-in-car-sales-cruise-past-1-5-million/>.
- [2] C. Lv, H. Wang, and D. Cao, "High-precision hydraulic pressure control based on linear pressure-drop modulation in valve critical equilibrium state," IEEE Trans. Ind. Electron., vol. 64, no. 10, pp. 7984–7993, Oct. 2017.
- [3] C. Lv, Y. Liu, X. Hu, H. Guo, D. Cao, and F. Y. Wang, "Simultaneous observation of hybrid states for cyber-physical systems: A case study of electric vehicle powertrain," IEEE Trans. Cybern., vol. PP, no. 99, pp. 1–11, Aug. 2017.
- [4] E. Musk, "Model S fire," Tesla Motors, San Carlos, CA, USA, Feb. 20, 2014.
- [5] "Charge your model S - adapter guide, high power charging, and supercharge," Tesla Motors, San Carlos, CA, USA, Jun. 23, 2012.
- [6] Tesla Motors: Battery Swap [Online]. Available: <http://www.teslamotors.com/batteryswap>.
- [7] "China Electric Vehicle Charging Station Market Report, 2012-2013," 2013, Research in China.
- [8] Mushfiqur R. Sarker, Hrvoje Pandžić and Miguel A. Ortega-Vazquez "Optimal Operation and Services Scheduling for an Electric Vehicle Battery Swapping Station", IEEE transactions on power systems, vol. 30, no. 2, march 2015.
- [9] Y. Guo, X. Liu, Y. Yan, N. Zhang, and W. Su, "Economic analysis of plug-in electric vehicle parking deck with dynamic pricing," in Proc. IEEE Power Energy Soc. Gener. Meeting, National Harbor, MD, USA, 2014, pp. 1–5.
- [10] Y. Guo, J. Xiong, S. Xu, and W. Su, "Two-stage economic operation of microgrid-like electric vehicle parking deck," IEEE Trans. Smart Grid, vol. 7, no. 3, pp. 1703–1712, May 2016.
- [11] Y. Zheng, Z. Y. Dong, Y. Xu, K. Meng, J. H. Zhao and J. Qiu, "Electric Vehicle Battery Charging/Swap Stations in Distribution Systems: Comparison Study and Optimal Planning," in IEEE Transactions on Power Systems, vol. 29, no. 1, pp. 221–229, Jan. 2014.
- [12] M. Cococcioni and L. Fiaschi, "The Big-M method with the numerical infinite M," Optimization Letters, 2021.
- [13] GUROBI Solver for mixed-integer conic programs accessed from <http://www.gurobi.com>.
- [14] YALMIP optimizer software accessed from <https://yalmip.github.io>.
- [15] Indian Energy Exchange Limited (IEX) power trading platform <http://www.iexindia.com/Reports/AreaPrice.aspx>.